

Amrita Nair

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Site Reliability Engineer with experience building and operating production systems on AWS and Azure, and a strong backend development background. Experience automating configuration changes and designing event-driven workflows with proper failure handling and observability. Comfortable owning systems end to end, participating in on-call, and improving reliability through automation. Interested in building reliable, scalable infrastructure and distributed systems.

Education

- 2021–2024 **Charles University**, Faculty of Mathematics and Physics(Matfyz), Czech Republic, Focus on transformer-based NLP models, including large language models, and applied ML systems.
MSc. Computer Science (Computational Linguistics — NLP, Large Language Models (LLM's))
- 2021–2024 **Saarland University**, Department of Language Science and Technology(LST), Germany.
MSc. Language Science and Technology
- 2015–2019 **Amrita School of Engineering**, Amrita Vishwa Vidyapeetham, Kerala.
Bachelors in Computer Science and Engineering

Experience

- August 2024 - Present **Associate Site Reliability Engineer**, TRICENTIS, Prague, Czech Republic.
- Designed and implemented a unified, event-driven synchronization system (S3 → SNS → Lambda) that consumes a JSON-based model configuration used by Terraform and automatically propagates model add/update/delete events across AWS and Azure via a single control path, updating MongoDB as the target datastore. Lambda function written in Python.
 - Owned the security design for cross-cloud model updates, using AWS KMS-backed signing to authenticate service-to-service API calls and avoid long-lived secrets or CI-managed private keys.
 - Engineered the system for production operations: idempotent update handling, CloudWatch observability, and DLQ-backed failure recovery.
 - Integrated an OpenTelemetry-based LLM observability platform by provisioning Kafka, PostgreSQL and enabling cross-region ClickHouse Cloud access via AWS PrivateLink, also replacing a bastion-host-based access model with a Lambda-based approach.
 - Refactored the dedicated Azure IaC repository for model deployments, modularizing Terraform code and reducing configuration size by ~90%, significantly improving reuse and long-term maintainability.
 - Performed a production Kubernetes upgrade by validating and upgrading cluster add-ons for compatibility.
 - On-call engineer using PagerDuty for escalations for a production AI platform, resolving incidents across Kubernetes, Terraform, AWS, and Azure.
- August 2019 - May 2021 **Backend Developer - Ruby on Rails**, BANG THE TABLE (NOW PART OF GRANICUS).
- Designed and implemented backend APIs in Ruby on Rails for a dashboard that allows users to embed and display third-party URLs, with extensive automated test coverage.
 - Maintained production services by diagnosing and fixing client and QA-reported issues related to data handling and display logic.
 - Improved code quality and reliability in a large Ruby on Rails codebase through refactoring and rigorous testing.

Projects

- 2026 **Clippy: Distributed Data Synchronization System.**
- Implemented CRDT data structures in Go (vector clocks, OR-Sets, LWW registers) with extensive tests. Created separate Go HTTP server with REST API and SQLite backend. (in-progress)

Interests

Python, Go, Terraform, AWS, Azure, Kubernetes, Distributed Systems